interferes with itself destructively when it gets there, ensuring that it never reaches detector 2. Thus, the photon hits only detector 1, and the world does not branch.

It doesn't matter whether the second beam splitter is put in place before or after the photon goes through the first beam splitter—all that matters is whether the second beam splitter is there when the wave function arrives.

Spontaneous-collapse theory: A photon wave function enters the first beam splitter, splits, and travels down both paths. If the second beam splitter isn't in place, the photon wave function hits both of the detectors and entangles with the detectors' wave functions. The enormous number of particles involved in this giant entangled wave function all but ensures that one of them will hit the collapse jackpot almost instantly, forcing the photon into one detector or the other entirely at random.

If the second beam splitter is in place, the photon wave function interferes with itself destructively when it gets there, ensuring that it never reaches the right-hand detector.

It doesn't matter whether the second beam splitter is put in place before or after the photon goes through the first beam splitter—all that matters is whether the second beam splitter is there when the wave function arrives.

In short, Wheeler's conclusions are, at best, not forced upon us. (At worst, they are logically incoherent.) Nor is this experiment particularly odd, seen from these other viewpoints—certainly not as strange as the Bell experiments. There are versions of this experiment that combine it with aspects of the Bell experiments, but those are similarly possible to explain in all of these interpretations (though the explanations do get quite a bit more complex).

One final note: although pilot waves are in general nonlocal, in this case, everything is entirely local using the pilot-wave interpretation. So there's a sense in which Wheeler was right—this is exactly the kind of thing that was at the heart of the Einstein-Bohr debates, because it can in theory be explained locally, yet adherents of Copenhagen insist on giving a nonlocal explanation for it!